

HE2003 Econometrics I: Linear Regression with One Regressor 2

Wenjie Wang

**Economics, School of Social Sciences
Nanyang Technological University**

Aug 26, 2024

Linear Regression with One Regressor

Example: The relationship between chief executive officer (CEO)'s salary and return on equity

In **finance**, the **return on equity (ROE)** is a very important measure of firm performance and is calculated as:

$$ROE = \frac{\text{Net income of the firm in a fiscal year}}{\text{Shareholder's investment (e.g., firm's stocks)}}.$$

The ROE measures a firm's performance by revealing **how much profit it can generate with the money invested**. ROEs of 15 – 20% are generally considered good. Companies such as **Google, Amazon, Facebook and Apple** can have ROEs as high as 30%.

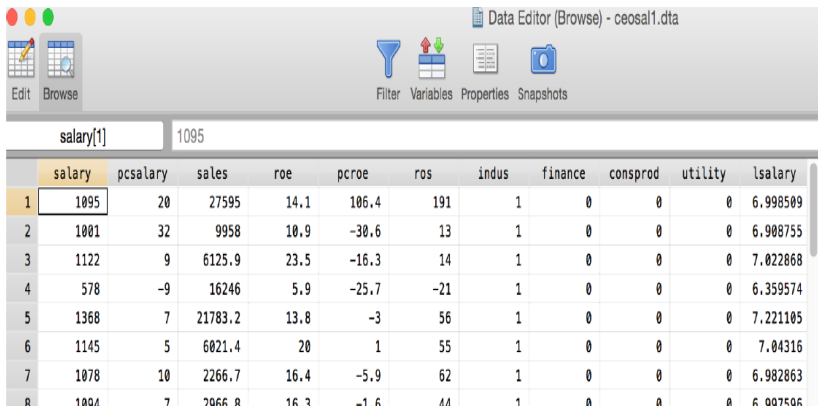
- To study the **effect** of firm performance (roe) on CEO's salary, we consider the following linear regression model with one regressor:

$$salary = \beta_0 + \beta_1 roe + u.$$

- The slope parameter β_1 measures the **change in the CEO's salary for a unit change in ROE** of his or her firm: $\beta_1 = \Delta salary / \Delta roe$.

Linear Regression with One Regressor

- We use the **dataset (ceosal1.dta)** with **209 CEOs in US** obtained from *Business Week*.
- In the data set, the variable **“salary”** is the annual salary of a CEO in the **unit of 1,000 US dollars** (Thus, 1095 = \$1,095,000), and the variable **“roe”** is the ROE of the CEO's firm in the **unit of %** (Thus, $roe = 10$ means that the ROE is equal to 10%).



Data Editor (Browse) - ceosal1.dta

salary[1] 1095

	salary	pcsalary	sales	roe	pcroe	ros	indus	finance	consprod	utility	lsalary
1	1095	20	27595	14.1	106.4	191	1	0	0	0	6.998509
2	1001	32	9958	10.9	-30.6	13	1	0	0	0	6.908755
3	1122	9	6125.9	23.5	-16.3	14	1	0	0	0	7.022868
4	578	-9	16246	5.9	-25.7	-21	1	0	0	0	6.359574
5	1368	7	21783.2	13.8	-3	56	1	0	0	0	7.221105
6	1145	5	6021.4	20	1	55	1	0	0	0	7.04316
7	1078	10	2266.7	16.4	-5.9	62	1	0	0	0	6.982863
8	1094	7	2066.8	16.3	-1.6	44	1	0	0	0	6.907506

Linear Regression with One Regressor

Estimation Result: The OLS regression line is

$$\widehat{salary} = 963.191 + 18.501roe, \quad n = 209$$

Question: How do we interpret this equation?

If the ROE increases by one percentage point, $\Delta roe = 1$, then CEOs' annual salary is predicted to change by about 18.5, or \$18,500.

Question: How do we compute the predicted values and residuals for each observation?

E.g., for the first observation, $roe_1 = 14.1$; We plug it into the OLS equation:

$$\widehat{salary}_1 = 963.191 + 18.501 \times 14.1 = 1224.05,$$

and $\widehat{salary}_2, \dots, \widehat{salary}_{209}$ can be computed similarly (This can be done in STATA using “**predict salary_hat**”). Then, the residual of the first observation

$$\hat{u}_1 = salary_1 - \widehat{salary}_1 = 1095 - 1224.05 = -129.05,$$

and $\hat{u}_2, \dots, \hat{u}_{209}$ can be computed similarly.

Linear Regression with One Regressor

Important Comparison: True vs Estimated

True (unknown) parameters: β_0, β_1

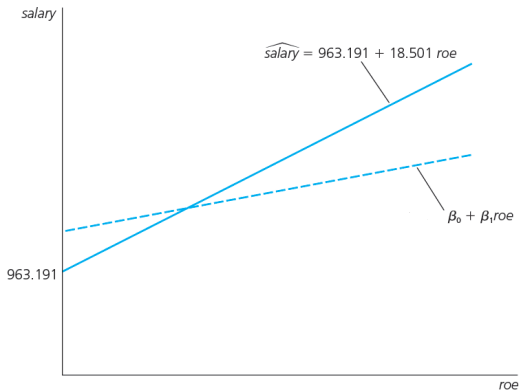
True (unknown) error term: u_i

True (unknown) pop.reg. line: $\beta_0 + \beta_1 X$

OLS estimators: $\hat{\beta}_0, \hat{\beta}_1$

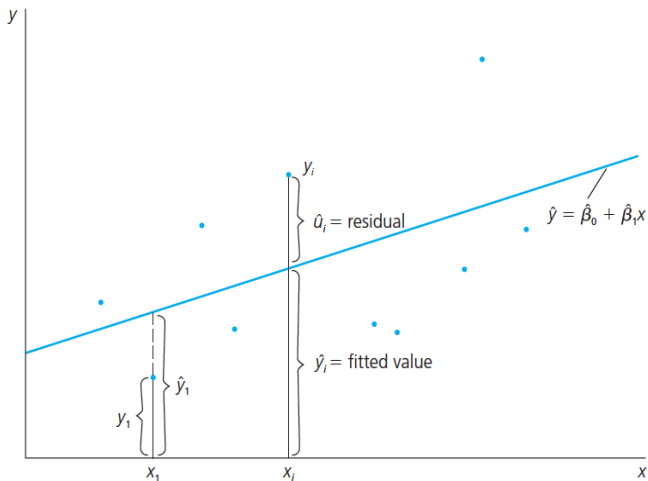
Residual: $\hat{u}_i$

OLS reg. line: $\hat{\beta}_0 + \hat{\beta}_1 X$



Linear Regression with One Regressor

Figure of predicted values (fitted values) and residuals



Linear Regression with One Regressor

KEY CONCEPT: The Least Square Assumptions

We need the following assumptions for the OLS to be a good estimator for β_0 and β_1 :

1. The error term u_i has conditional mean zero given X_i : $E[u_i|X_i] = 0$;
2. (X_i, Y_i) , $i = 1, \dots, n$ are independent and identically distributed (i.i.d.);

and

3. Large outliers of data are unlikely.

Initially, these assumptions may appear abstract. They do, however, have natural interpretations, and understanding these assumptions is essential for understanding when the OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ will - and will not - give useful estimates of β_0 and β_1 .

Linear Regression with One Regressor

Assumption 1: The error term u_i has conditional mean (conditional expectation) zero given X_i : $E[u_i|X_i] = 0$.

- Assumption 1 is about the “other factors” (the error term u).
- It asserts that these other factors are not affected by the explanatory variable X in the following sense:
For any given value of X , the mean of u has to be zero (Zero Conditional Mean).
- Intuitively, to estimate the causal effect of X on Y , we need to somehow “separate” the explanatory variable X and the error term u .
- Note: $E(u|X) = 0$ implies that $E(u) = 0$ and $\text{Corr}(u, X) = 0$ (by the Law of Iterated Expectations (L.I.E.); The details of L.I.E. are not required in this course). This means $E(u|X) = 0$ is the strongest among the three.

Linear Regression with One Regressor

Assumption 2: (X_i, Y_i) , $i = 1, \dots, n$ Are Independently and Identically Distributed

This is satisfied naturally if the individual (or district, etc.) is sampled by **simple random sampling**:

- The individuals are selected from the **same population**, so that they are **identically distributed**.
- The individuals are selected completely **at random**, so that they are **independently distributed**.

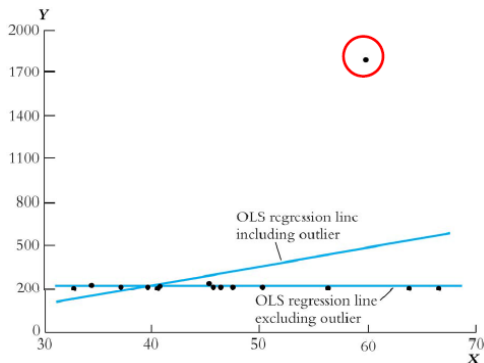
For instance, let X be the **age** of a worker and Y be his or her **earnings**, and imagine **drawing at random from the population of all workers** (e.g., all workers in Singapore).

If **a sample of n workers** is drawn only from this population, then (X_i, Y_i) , $i = 1, \dots, n$ necessarily have the **same distribution**. By contrast, if some are drawn from the US population, then (X_i, Y_i) , $i = 1, \dots, n$ may not have the same distribution.

Linear Regression with One Regressor

Assumption 3: Large Outliers Are Unlikely

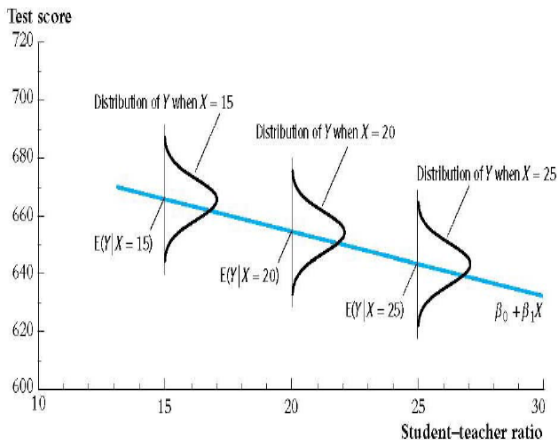
This assumption limits the cases of drawing an observation with **extremely large value** of X_i or Y_i , because the **outliers** would be given great importance in an OLS regression (remember the square in the OLS minimization problem), and would **make the OLS regression results misleading**.



Linear Regression with One Regressor

More Explanation on the Least Square Assumption 1: $E[u_i|X_i] = 0$ (or can be written as $E[u|X] = 0$).

Assumption 1 is illustrated in the figure. At any given value of X ($X = 15, 20, 25, \dots$), (i) Y is distributed around the population regression line, and (ii) the error, $u = Y - (\beta_0 + \beta_1 X)$, has a mean of zero.



Linear Regression with One Regressor

- In our example, the **population regression line** is the **relationship that holds on average** between class sizes (STRs) and test scores, and the **error term** u_i represents the other factors that lead test scores at the school district i to **differ from the prediction based on the population regression line** (deviation).
- As shown in the figure, **at a given value** of X (STR in this example), say 20 students per teacher, **for some school districts** these other factors lead to better performance than predicted ($u_i > 0$, and Y_i **above the line**), **for other districts** they lead to worse performance ($u_i < 0$, and Y_i **below the line**), but **on average the prediction** based on the population regression line **is right**.
- In other words, **given $X = 20$, the mean of the error term is zero**. And this **holds for all the other values of X** as well.
 - $E[u_i|X_i = 20] = 0, E[u_i|X_i = 15] = 0, E[u_i|X_i = 25] = 0, \dots$
 - This can be written compactly as: $E[u_i|X_i] = 0$.

Linear Regression with One Regressor

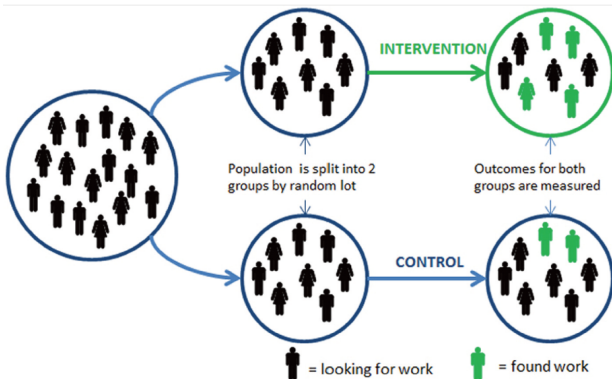
A useful **benchmark** for thinking about Assumption 1 is to consider an **ideal randomized experiment**:

- In a randomized experiment, the explanatory variable X (certain treatment or intervention) is **randomly assigned** (e.g., patients randomly assigned to medical treatments).
- Because X is assigned randomly, **all other individual characteristics** (things that are in the error term u) are totally independent from X ; That is to say, **u and X are totally independent**. This immediately implies that $E[u|X] = 0$ because of the **property of conditional expectation**. Therefore, for randomized experiments we think that **Assumption 1 typically holds**.
- On the other hand, with **observational data**, we will **need to be careful about** about whether $E[u|X] = 0$ holds.

Linear Regression with One Regressor

The random assignment makes the **treatment and control groups very similar**, so that **comparing the outcome** of two groups gives us the “**causal effect**” of the program/intervention.

Question: What does u represent here?



Linear Regression with One Regressor

Example (A randomized controlled experiment)

In a randomized experiment, the subjects i are randomly assigned to the treatment group (e.g., take part in some job training program) and control group (not take part in the training program); This also can be written as a linear regression model:

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, \dots, n.$$

where Y_i is certain outcome (e.g., labor market outcome: employed or unemployed) and X_i is the group indicator or treatment indicator with $X_i = 1$ or $X_i = 0$:

$$\begin{aligned} X_i &= 1 && \text{if individual } i \text{ is assigned to the treatment group,} \\ X_i &= 0 && \text{if individual } i \text{ is assigned to the control group.} \end{aligned}$$

So we want to explain the outcome Y_i by whether subject i received certain treatment or not.

Q: What is the meaning of β_0 and β_1 here? (Hint: We can use the conditional expectation to find out)

Linear Regression with One Regressor

Example (A randomized controlled experiment (cont'd))

The random assignment typically is done using a **computer program**.

Therefore, X (the treatment indicator, which is decided by the computer) is **independent from all personal characteristics** of the subject (years of education, ability, experience, etc.), which are **inside the error term u** .

That is to say, in a randomized controlled experiment, the **random assignment makes X and u independent** (full independence), which in turn implies that Assumption 1, the **zero conditional mean assumption**, holds in the case of randomized experiment:

$$\text{Full Independence} \rightarrow E[u|X] = 0.$$

This is because 1) Full independence between X and u implies that $E[u|X] = E[u]$, and 2) we can always assume that $E[u] = 0$.

Linear Regression with One Regressor

Example (Return to education: possible violation of Assumption 1)

For observational data, we **need to think hard** whether Assumption 1 holds. For instance, suppose we are interested in the so-called “**return to education**”: If given another year of education, by how much will the wage increase?

$$wage = \beta_0 + \beta_1 educ + u$$

Now, let us assume that u is equal to “**innate ability**” ($u = abil$).

Note that Assumption 1 requires that the **average level of ability is the same** regardless of years of education. For example, it requires:

$$E[abil|educ = 10] = E[abil|educ = 16],$$

where $E[abil|educ = 10]$: average ability of those who have **10 years of educ.** $E[abil|educ = 16]$: average ability of those who have **16 years of educ.**

Linear Regression with One Regressor

Example (Return to education (cont'd))

In fact, it assumes that the average ability is the **same for all education levels**: $\dots = E[abil|educ = 8] = E[abil|educ = 9] = E[abil|educ = 10] = E[abil|educ = 11] = E[abil|educ = 12] = \dots$

If we think that **average ability increases with years of education** (which may be quite true), then Assumption 1 is **false**:

$$E[abil|educ = 10] < E[abil|educ = 16].$$

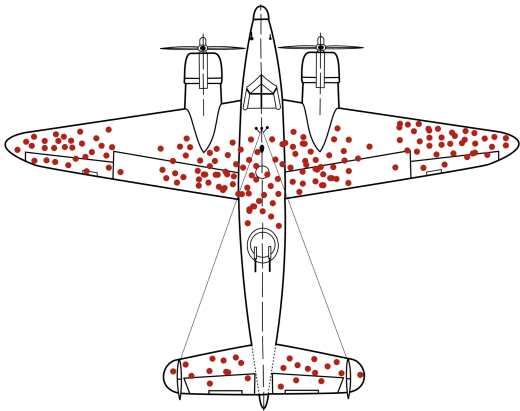
For example, this would happen if, on average, people with **higher ability choose to be more educated** (e.g., choose to pursue an MBA for further career development), or **higher innate ability leads to higher education** (e.g., to enter universities, one has to pass the entrance exams first; this is called “**selection bias**” or “**survivorship bias**”). In this case, “educ” and “abil” will be dependent with each other.

Q: How about the model: $\text{Test Score} = \beta_0 + \beta_1 \text{STR} + u?$

Linear Regression with One Regressor

Example (Survivorship Bias)

During World War II, the famous statistician **Abraham Wald** at Columbia University was requested by the US Navy to study **how to minimize bomber losses to enemy fire** (red points in the figure represent the damaged areas from bomber samples).



Linear Regression with One Regressor

Example (Survivorship Bias (cont'd))

- Wald examined the damage done to aircraft that had returned from missions and recommended adding armor to the areas that showed the least damage.
- The bullet holes in the returning aircraft represented areas where a bomber could take damage and still fly well enough to return safely to base.
- Therefore, Wald proposed that the Navy reinforce areas where the returning aircraft were unscathed, inferring that planes hit in those areas were the ones most likely to be lost.
- This is an example of survivorship bias because crucial data from fatally damaged planes was being ignored; those hit in other places did not survive.
- Wald's work is considered seminal in the then nascent discipline of operational research.